Wafer Map Anomaly Detection

A Comparative Study of Deep Learning Approaches

50.039 Deep Learning · Y2026

Group 8

The Problem

Semiconductor wafers must be inspected for defects before chips are shipped.

- **Defects cause yield loss** — each faulty chip costs money and delays production
- **Manual inspection is slow** — 811k wafer maps cannot be reviewed by hand
- **Labelled defect data is scarce** — in production, defects are rare and often unlabelled
- **9 distinct defect patterns** — scratch, edge-ring, donut, center, loc, edge-loc, random, near-full, and none

> **Goal:** Build a system that detects defective wafers automatically, using only normal wafers for training.

The Dataset: WM-811K

811,457 wafer maps from the LSWMD dataset

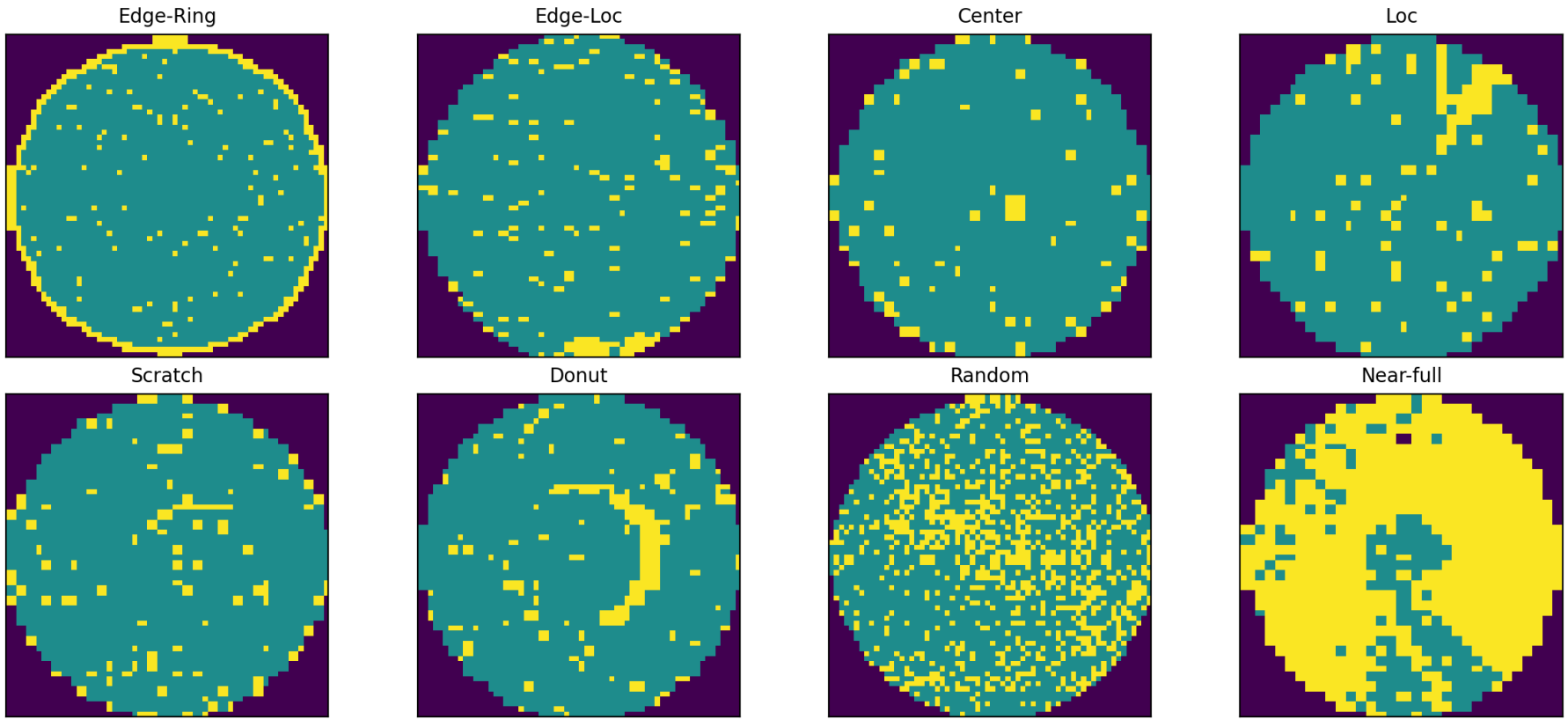
Split	Count	Share
Normal (labelled)	147,431	18.2%
Defective (labelled)	25,118	3.1%
Unlabelled	638,908	78.7%

Key challenge: Class imbalance — a model that predicts every wafer as normal achieves 96%+ accuracy while catching zero defects.

The 9 defect pattern types differ in shape, size, and sparsity — making detection non-trivial.

Defect Type Gallery

Representative Defect Patterns from WM-811K Test Set



Our Plan: Normal-Only Training

We train **exclusively on normal wafer maps**. At inference, each wafer receives an anomaly score — higher means more deviant from normal.

Why not supervised classification?

- Requires balanced, labelled defect data — scarce in production
- Cannot detect defect types unseen at training time
- Anomaly detection generalises to new defect patterns automatically

Training regime:

- **Train:** normal wafers only
- **Validation:** normal wafers only (threshold calibration)
- **Test:** normal + all labelled defect types (held out completely)

How We Evaluate

Three metrics — because **accuracy alone is useless** for imbalanced data.

Metric	What it measures	Why it matters here
AUROC	Ranking quality across all thresholds	Threshold-free; our primary metric
AUPRC	Precision-recall area under curve	Sensitive to imbalanced class performance
F1	Harmonic mean at the deployed threshold	Reflects real production performance

A model that predicts all wafers as normal achieves **96%+ accuracy** and **zero defect recall**. AUROC and AUPRC expose this immediately.

Thresholding Strategy

We set the anomaly threshold **without using any defect labels** — simulating deployment.

Strategy: 95th percentile of validation anomaly scores

1. Run the trained model on the normal-only validation set
2. Record all anomaly scores
3. Set the threshold at the 95th percentile — 5% of normals will be flagged as false positives
4. Apply this threshold unchanged to the held-out test set

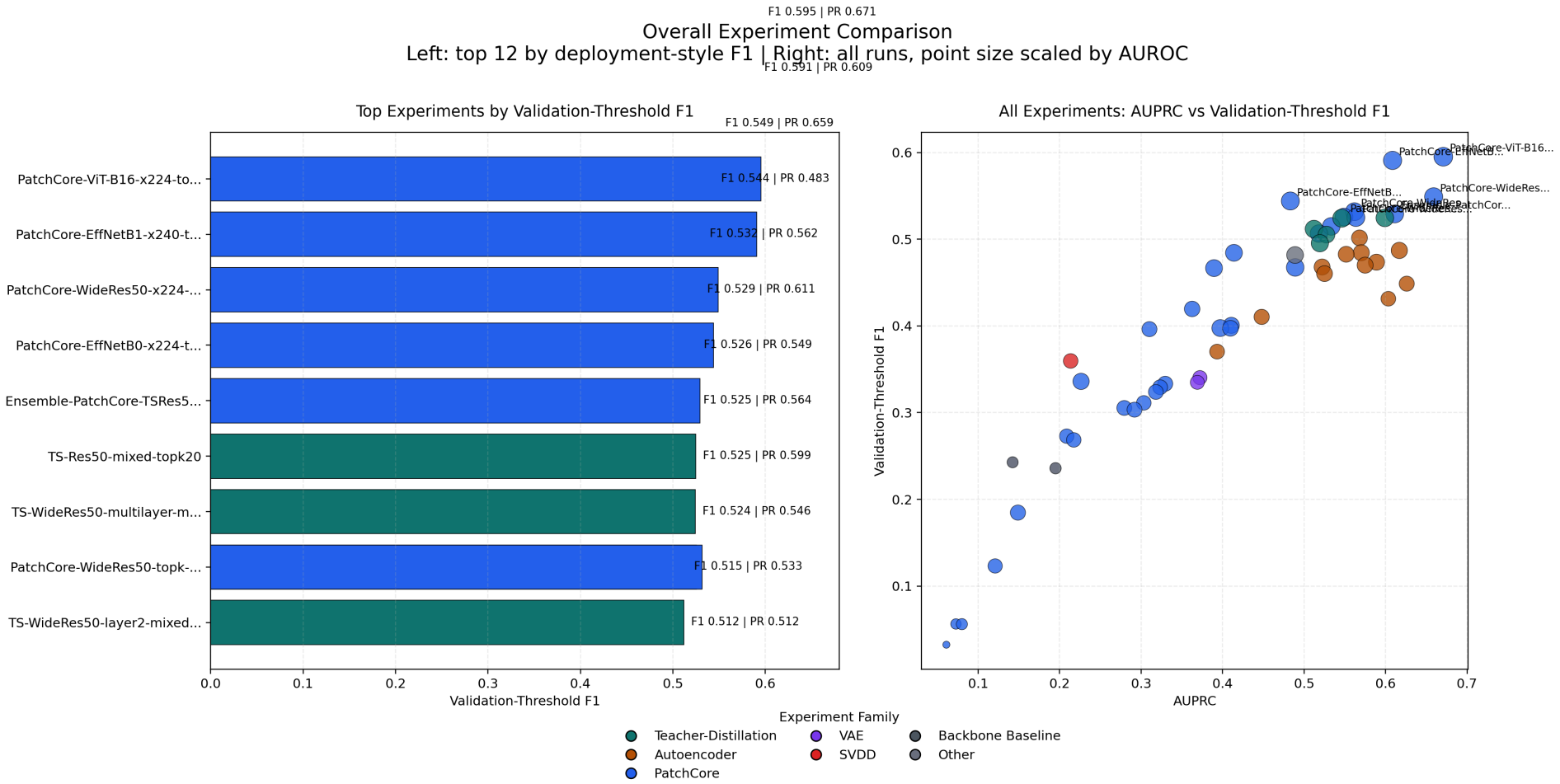
> This is a realistic constraint: in production, you calibrate on what you have — normal wafers — and accept a fixed false positive rate as the operating point.

Top 10 Models — Where We Ended Up

Rank	Model	F1	AUROC	AUPRC
1	**Ensemble: ViT-B/16 + DINOv2 (max-fusion)**	**0.623**	**0.967**	0.716
2	PatchCore + ViT-B/16, MAE fine-tuned (25% mask)	0.594	0.962	**0.762**
3	PatchCore + ViT-B/16, MAE fine-tuned (75% mask)	0.595	0.959	0.717
4	PatchCore + ViT-B/16, frozen, 224×224	0.595	0.956	0.671
5	PatchCore + EfficientNet-B1, 240×240	0.591	0.935	0.609
6	PatchCore + WideResNet50-2, 224×224	0.549	0.931	0.659
7	Teacher-Student + ResNet50	0.525	0.909	0.599
8	Teacher-Student + WideResNet50-2	0.524	0.923	0.546
9	Autoencoder (plain, 224×224)	0.510	0.901	0.596
10	WideResNet50-2 center-distance	0.243	0.677	0.142

The journey from row 10 to row 1 is what the next section explains.

Overall Model Comparison



Method Journey: Six Hypotheses

We tested six questions in sequence — each answer motivated the next question.

	Hypothesis	Key result
H1	Can reconstruction error detect defects?	AE works but hits a ceiling
H2	Do pretrained global features do better?	Global pooling discards spatial info
H3	Do local patch features matter?	**Large jump** — local scoring is the key
H4	Does backbone architecture matter?	ViT-B/16 at 224×224 wins
H5	Does MAE fine-tuning the ViT help?	AUPRC improves; F1 stays stable
H6	Can ensembles push further?	+0.011 AUROC with ViT + DINOv2

H1: Can Reconstruction Error Detect Defects?

Setup: Train an Autoencoder and Beta-VAE on normal wafers. Score each test wafer by its reconstruction error — defective wafers should reconstruct poorly.

Architecture: Convolutional encoder-decoder, MSE loss, 64×64 input.

Hypothesis: Defective wafers are out-of-distribution — the AE will fail to reconstruct them, producing high reconstruction error.

H1 Result: Works, But Has a Ceiling

Best result: Autoencoder — F1 = 0.510 · AUROC = 0.901

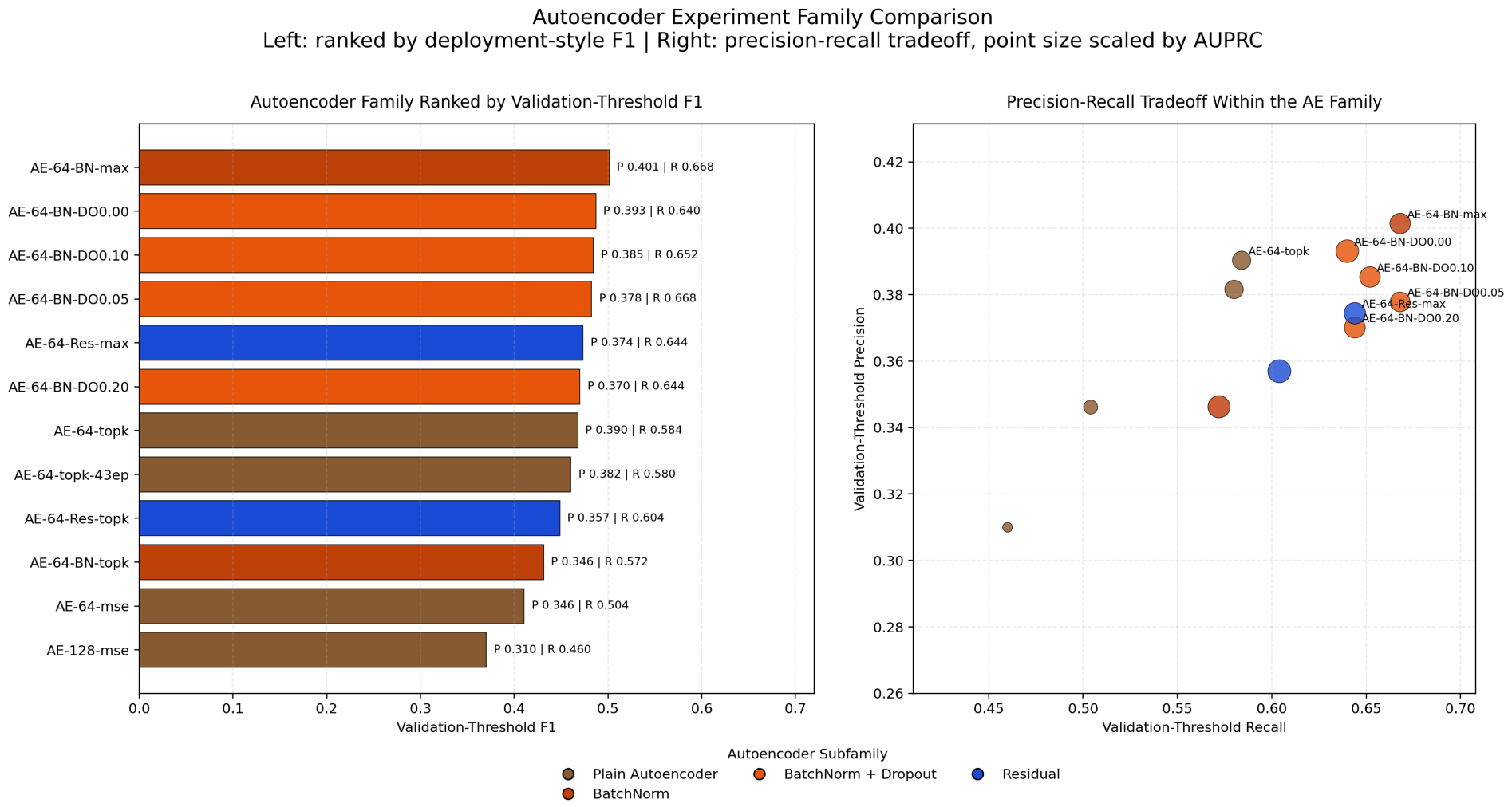
What worked: The AE does assign higher scores to defectives on average. A viable baseline.

Key limitation: The AE also learns to reconstruct defects — they are not out-of-distribution enough for a pixel-level model.

- Edge-ring and near-full defects score high ✓
- Scratch and loc defects are nearly invisible — sparse linear defects do not raise reconstruction error meaningfully ✗

> **Conclusion:** Reconstruction is a weak anomaly signal. We need features that represent what normal looks like with more precision than pixel values.

H1: Autoencoder Family Score Distributions



H2: Do Pretrained Global Features Do Better?

Setup: Freeze a pretrained backbone (ResNet18, WideResNet50-2). At inference, compute the cosine distance from each wafer's global embedding to the centroid of normal embeddings.

Hypothesis: ImageNet features encode richer visual semantics than pixel-level reconstruction error, and will better separate normal from defective.

H2 Result: Global Pooling Discards Too Much

WideResNet50-2 centroid distance: F1 = 0.243 · AUROC = 0.677

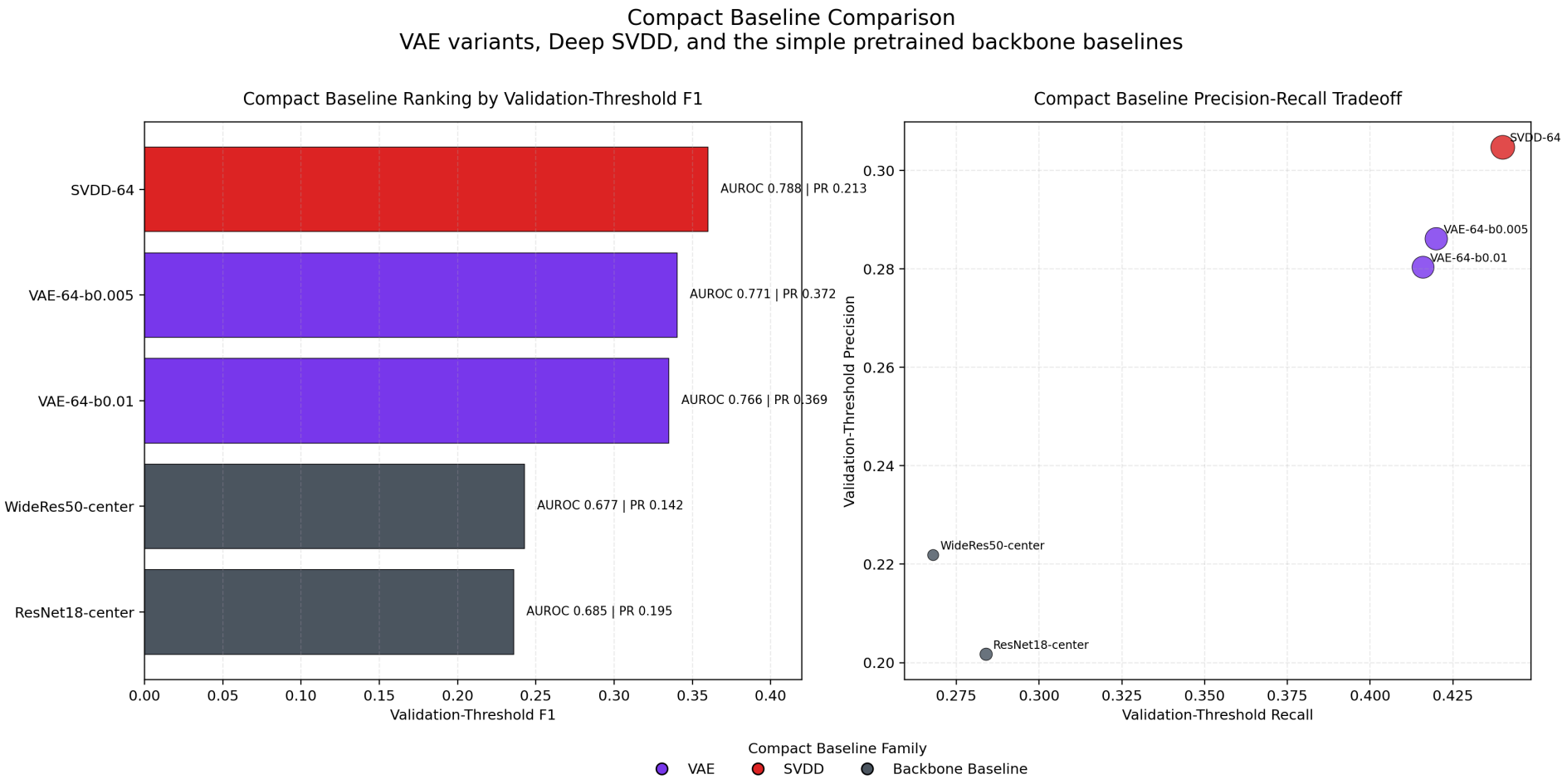
Worse than the AE. Global average pooling collapses all spatial structure into a single vector — a defect occupying 5% of the wafer disappears.

Teacher-Student distillation (ResNet50): F1 = 0.525 · AUROC = 0.909

Better — multi-layer distillation preserves more spatial structure than raw centroid distance, improving over the AE.

> **Key insight:** The failure is not the pretrained features — it is the pooling. Spatial structure matters. We need to score at the **patch level**, not the image level.

H2: Global Feature Baseline Comparison



H3: Do Local Patch Features Matter?

Setup: PatchCore — extract patch-level features from a frozen pretrained backbone, build a memory bank of normal patches, score each test wafer by its maximum nearest-neighbour distance to the memory bank.

Key idea: Instead of comparing entire images, compare individual patches. A defect occupying 5% of the wafer will be caught if even one patch is anomalous.

Hypothesis: Patch-level scoring will dramatically outperform global image scoring.

H3 Result: Local Features Are the Key Insight

PatchCore + WideResNet50-2: F1 = 0.549 · AUROC = 0.931

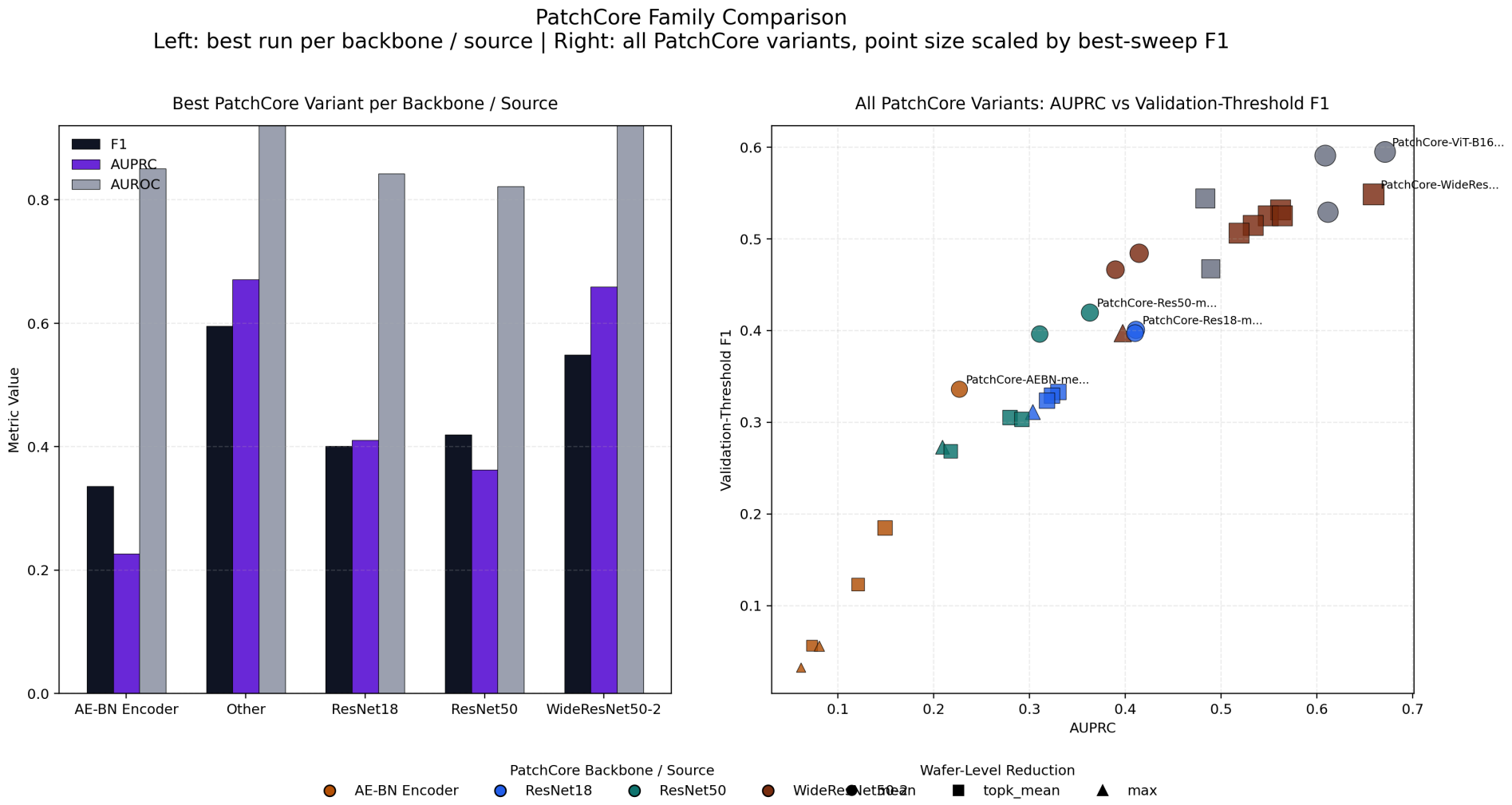
A large jump — from AUROC 0.909 (Teacher-Student) to 0.931 in one step.

What changed: The granularity of comparison, not the backbone. The same WideResNet50 features that failed at the image level succeed at the patch level.

- Scratch patterns — nearly invisible to the AE — are now detectable ✓
- Edge-ring patterns, already well-detected, improve further ✓

> **Conclusion:** PatchCore is the right framework. The next question: which backbone produces the best patch features?

H3: PatchCore Family Comparison



H4: Does the Backbone Architecture Matter?

Setup: Fix the PatchCore framework. Sweep backbone architectures and input resolutions with all other hyperparameters held constant.

Backbones: ResNet18 · ResNet50 · WideResNet50-2 · EfficientNet-B0/B1 · ViT-B/16 · DINOv2 ViT-B/14

Resolutions: 64 × 64 · 128 × 128 · 224 × 224 · 240 × 240

Hypothesis: Stronger backbone features and higher resolution will improve anomaly scores.

H4 Result: ViT-B/16 at 224×224 Wins

Backbone	Resolution	F1	AUROC
WideResNet50-2	64×64	0.532	0.917
WideResNet50-2	224×224	0.549	0.931
EfficientNet-B1	240×240	0.591	0.935
ViT-B/16 (frozen)	**224×224**	**0.595**	**0.956**

Why ViT? Transformer self-attention captures long-range spatial relationships across the entire wafer — context that CNN local receptive fields cannot see. Combined with PatchCore's local scoring, ViT provides both global context and local precision simultaneously.

H5: Does MAE Fine-tuning the ViT Improve Results?

Setup: Apply Masked Autoencoding (MAE) pre-training to ViT-B/16 using normal wafer maps before running PatchCore. The ViT learns to reconstruct randomly masked patches — adapting ImageNet features to the wafer domain.

Two masking ratios tested: 75% (aggressive masking) and 25% (light masking)

All PatchCore hyperparameters are identical to the frozen ViT-B/16 baseline.

Hypothesis: Domain-adapted features will improve anomaly scoring above ImageNet-pretrained features alone.

H5 Result: AUPRC Improves Substantially; Deployed F1 Stays Stable

Backbone	F1	AUROC	AUPRC	Recall
Frozen ViT-B/16 (ImageNet)	0.595	0.956	0.671	~80%
MAE fine-tuned (mask 75%)	0.595	0.959	0.717	82.4%
MAE fine-tuned (mask 25%)	**0.594**	**0.962**	**0.762**	**84.4%**

Why 25% masking wins: Wafer maps are sparse binary grids. At 75% masking, only 49 of 196 patches remain visible — too little context. At 25%, 147 patches remain, giving the model enough local structure to learn meaningful wafer representations.

Why F1 is stable: The threshold is fixed at 5% FPR on validation normals. AUPRC (+0.091) and recall (+4.4%) reveal real improvements that the deployed F1 cannot capture.

H6: Can Ensembles Push Further?

Setup: Combine anomaly scores from two complementary backbones — ViT-B/16 (ImageNet supervised) and DINOv2 ViT-B/14 (self-supervised, contrastive). Fusion strategy: take the **maximum score** per wafer across both models.

Hypothesis: Two backbones with different pretraining objectives will catch complementary defect patterns, yielding better coverage than either alone.

H6 Result: Ensemble Reaches Best Overall Performance

Model	F1	AUROC	AUPRC
PatchCore + ViT-B/16 (frozen)	0.595	0.956	0.671
PatchCore + DINOv2 ViT-B/14	—	~0.950	—
Ensemble: ViT-B/16 + DINOv2 (max-fusion)	**0.623**	**0.967**	**0.716**

Gain over single best model: +0.011 AUROC · +0.028 F1 · +0.045 AUPRC

Observation: The gain is consistent but modest. Backbone diversity (different pretraining objectives) matters more than simply doubling model count. The remaining gap reflects domain difficulty — sparse wafer patterns — not model capacity.

What If: What If We Had Used Defect Samples in Training?

Our entire approach assumes **normal-only training**. But what if that assumption costs us performance?

1. Defect-tuning PatchCore: Tune selected ViT-B/16 backbone layers using contrastive loss with defect examples alongside normals.

2. Supervised classifier: Train a classification model using 1,016 labelled defect samples across all nine defect classes.

> **The question:** Does knowing what defects look like actually help?

What If Result: Normal-Only Is the Right Choice

Defect-tuning results (PatchCore ViT-B/16):

Variant	F1	AUROC
Normal-only — baseline	0.595	0.956
One-layer defect tuning	0.607	0.957
Two-block defect tuning	**0.383**	0.932

One-layer barely gains (+0.012 F1). Two-block **catastrophically collapses** — defect supervision disrupts the learned normal distribution.

Supervised classifier: Matched AUROC = 0.956, but **Scratch recall = 0.167** vs **0.727** for normal-only PatchCore. It cannot detect defect types underrepresented at training time.

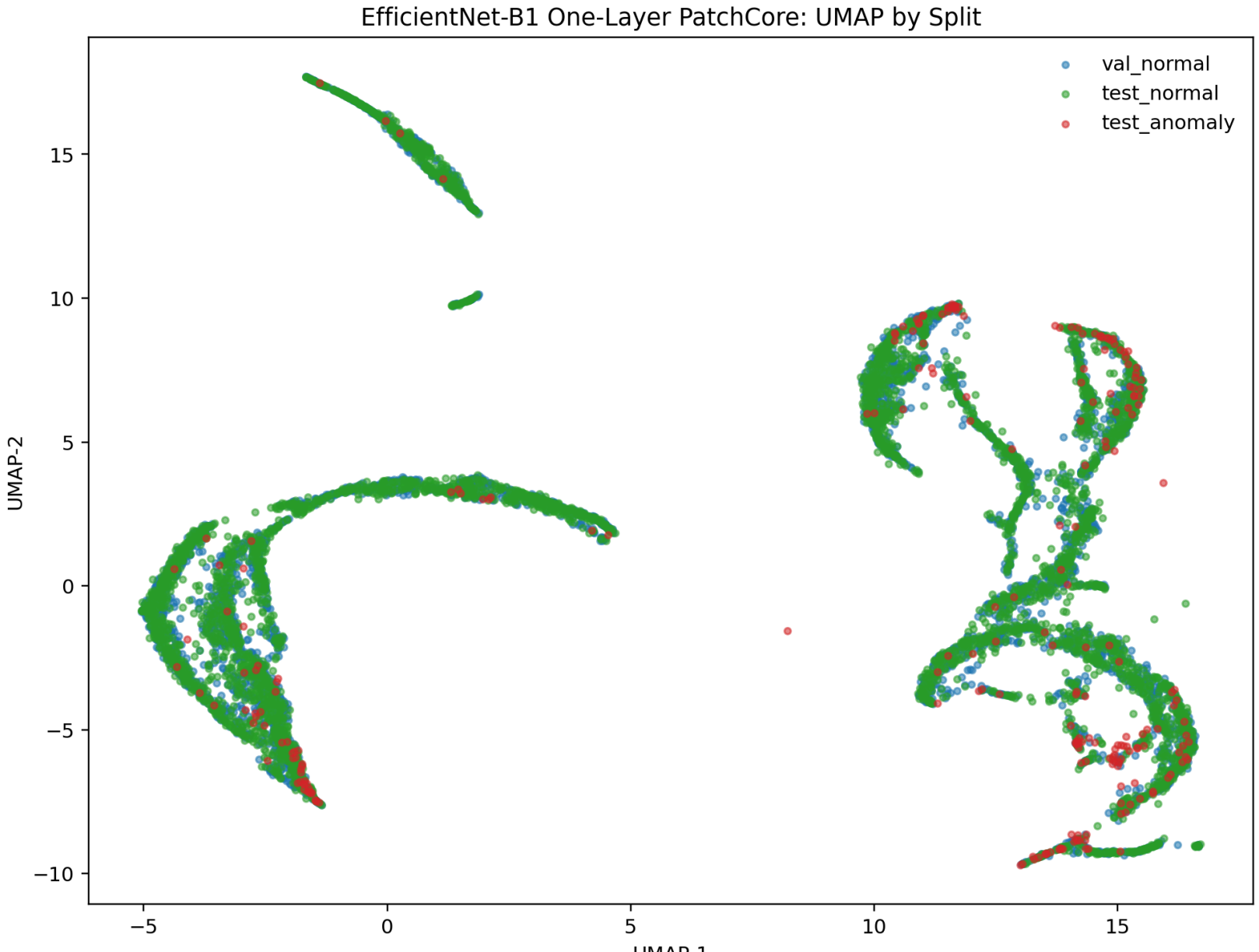
> **Conclusion:** Normal-only anomaly detection is not just a practical constraint — it is the more robust choice.

Final Results vs Published Baselines

Method	F1	AUROC
Jo & Lee (2021) — best published on WM-811K	< 0.50	—
Our AE baseline	0.510	0.901
Our best single model (PatchCore + ViT-B/16)	0.595	0.956
Our best ensemble (ViT-B/16 + DINOv2)	**0.623**	**0.967**
Roth et al. (2022) PatchCore on MVTec AD	—	~0.990

Our results substantially exceed prior WM-811K work (Jo & Lee 2021). The remaining gap vs MVTec AD reflects domain difficulty — sparse wafer patterns and strict deployment-style thresholding — not implementation quality.

What PatchCore Learned: Feature Space Separation



Key Takeaways

What the experiments showed:

- **Local > global:** Patch-level PatchCore outperforms global reconstruction and global embedding distance
- **Pretrained > from scratch:** Frozen ImageNet ViT features are strong; MAE domain adaptation further improves precision-recall (+0.091 AUPRC)
- **ViT > CNN:** Transformer attention captures long-range wafer context that CNN receptive fields miss
- **Ensembles add consistent gains:** Diverse pretraining objectives (supervised + self-supervised) complement each other
- **Normal-only is the robust choice:** Defect supervision either barely helps or catastrophically hurts; supervised models fail on unseen defect types

Limitation: Threshold calibration without defect labels is imprecise — F1 understates true model quality. AUPRC is the more informative signal

Thank You

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Member	Contribution
Member 1	[Area]
Member 2	[Area]
Member 3	[Area]

Questions?